

From Bed Rest to Mars:

Development of a Literature-Derived Machine Learning Framework for Predicting Lower-Limb Muscle Atrophy in Spaceflight Analogues

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DGLRM 2026

Agenda

Part 1: background and data

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The marker in the bottom-right corner of each slide shows where we are.

Why forecast muscle loss?

PLACEHOLDER: content by Niloufar

Suggested points from the final report:

- Unloaded muscles that hold the body up against gravity lose mass and strength
- A transit to Mars takes about six months; the crew must stand, move and work on arrival
- Countermeasures are planned before departure, so planners need to know how much of a given muscle is lost after a given number of days

From bed rest to Mars: why use bed-rest data?

PLACEHOLDER: content by Niloufar

Suggested points from the final report:

- Spaceflight data on leg muscle size are scarce, and the groups measured are small
- Bed rest is the established ground model: healthy volunteers lie in bed, usually tilted 6° head-down so that body fluids shift towards the head as in orbit
- Three decades of campaigns at NASA, MEDES (Toulouse), Charité (Berlin), DLR (Cologne) and elsewhere
- Bed rest imitates weightlessness but is not the same

What range of unloading did we study?

PLACEHOLDER: content by Niloufar

Suggested points from the final report:

- Unloading models: head-down and horizontal bed rest, dry immersion, one-leg suspension; spaceflight kept separate
- At least 5 days of continuous unloading; scans from day 5 to day 119
- Leg muscle size measured as volume, cross-sectional area, thickness or lean mass, by MRI, CT, DXA, pQCT or ultrasound

How the dataset was built

PLACEHOLDER: content by Niloufar

Suggested points from the final report:

- Systematic search of PubMed, Scopus, Web of Science and NASA's technical reports, 2013 onwards: 5,731 records
- Plus 9 older studies and one campaign from NASA's open bed-rest archive
- Extraction form fixed before extraction: one row per study, group, muscle and scan day
- Papers that report the same volunteers merged into one campaign

Possible figure: search flow diagram (figures/F1_corpus)

The final dataset

PLACEHOLDER: content by Niloufar

Suggested points from the final report:

- 742 measurements from 52 studies (1992 to 2026) and 36 independent campaigns
- Mostly MRI volumes of named muscles in young men during head-down bed rest
- Uneven: the MEDES 90-day campaign gives 40% of rows; 15 campaigns give 5 rows or fewer
- Used for modelling: 346 measurements from 32 campaigns

Possible figure: campaign timeline (figures/F1b_campaigns)

What was extracted from each study

PLACEHOLDER: content by Niloufar

Suggested points from the final report:

- Muscle, day of the scan, planned length of bed rest, % change from before bed rest
- Group: control or countermeasure, group size, sex, age
- Measurement: imaging method, quantity, position along the muscle
- Source of every value: DOI and page, table or figure

Are all muscles affected the same?

PLACEHOLDER: content by Niloufar

Suggested points from the final report:

- No. Calf muscles (plantar flexors) lose most: –15.8% at day 60, 5.7 pp more than the front thigh
- Front thigh, back thigh and shin muscles lose around 10%; hip muscles least
- 30% of the differences between measurements lie between muscles within the same study
- Groups with countermeasures lose 3.7 pp less than controls

Possible figure: muscle ranking (figures/F3_muscles)

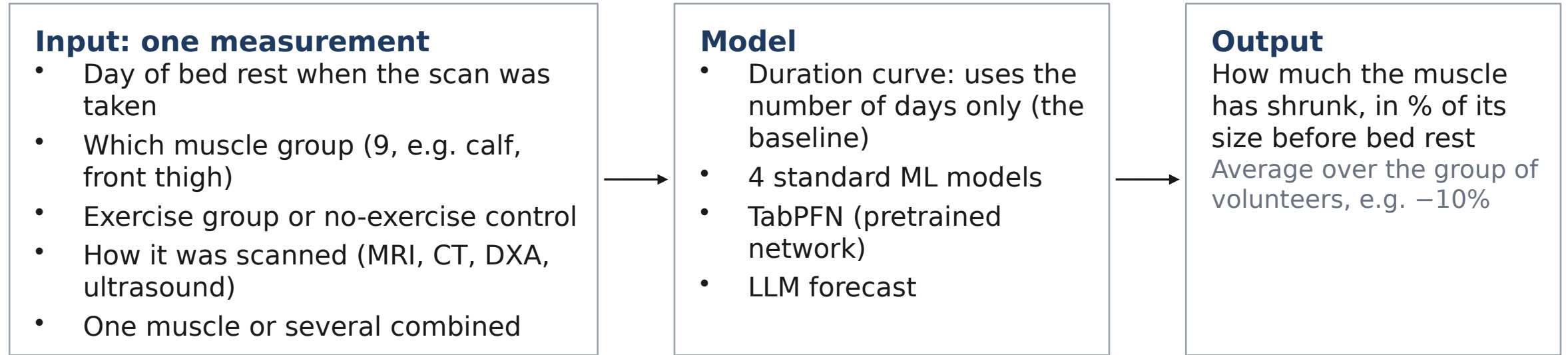
From data to forecasts

PLACEHOLDER: content by Niloufar

Suggested points from the final report:

- Bridge to part 2: the data show that loss depends on time and on the muscle
- Next question: can a model forecast the loss in a study it has never seen?

Every model is tested on a study it has not seen



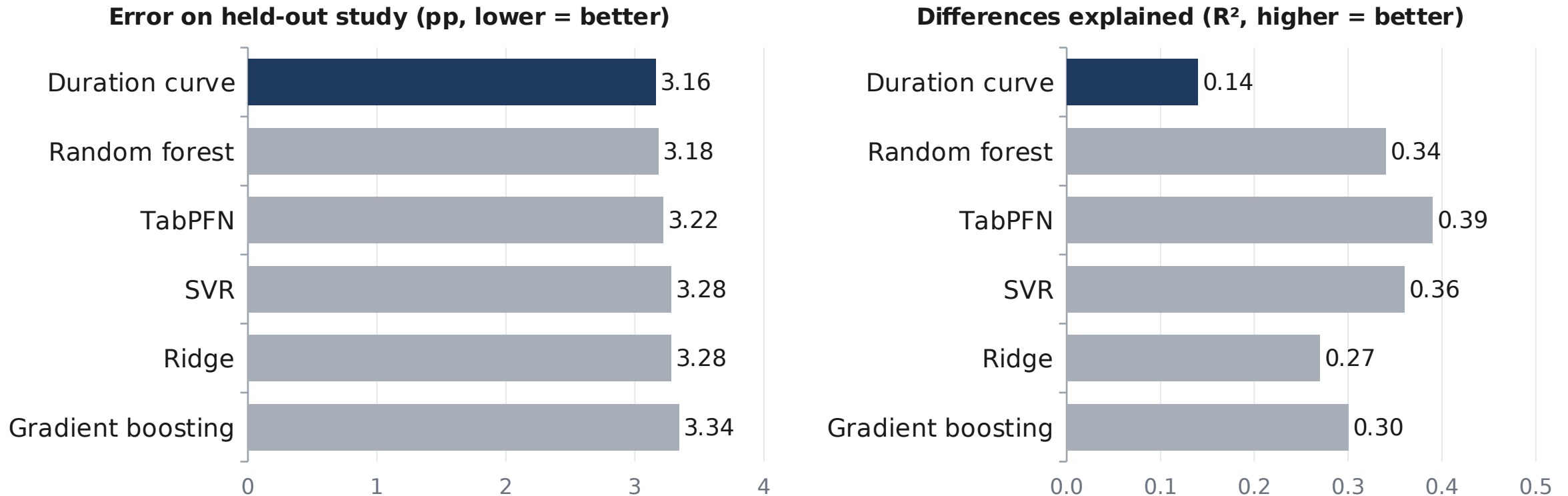
Testing: hold out one study, train on the rest, repeat 32 times



Grey: 31 training studies. Red: the held-out test study. Each study is held out once.

- Why studies, not measurements: the 346 measurements come from only 32 independent studies, because several papers report the same volunteers.
- Score: average error in percentage points (pp). Forecast -10%, measured -13%: error 3 pp. Each study counts once.

No ML model has lower error than the duration curve



- R^2 is the share of the differences between measurements that a model explains: 0 is none, 1 is all. The ML models explain more (0.27 to 0.39 vs 0.14) because they learn which muscles shrink more.
- Their error on a new study is still no lower. Most of the error left is a shift that affects a whole study (its scanner, volunteers, protocol), and none of our inputs describe it.

LLM forecast: what is sent and what comes back

Request (one per measurement)

participants

men only, mean age 35, n = 11

protocol

head-down tilt bed rest, 21 days, control group

measurement

vastus lateralis, MRI cross-sectional area

target

day 21 of bed rest

typical_curve_other_campaigns

duration curve fitted on the other 31 studies

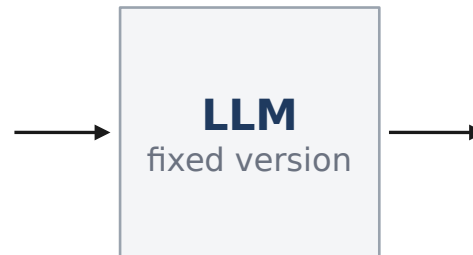
observations_other_campaigns

rows from the other 31 studies, most similar first, up to about 26,000 tokens

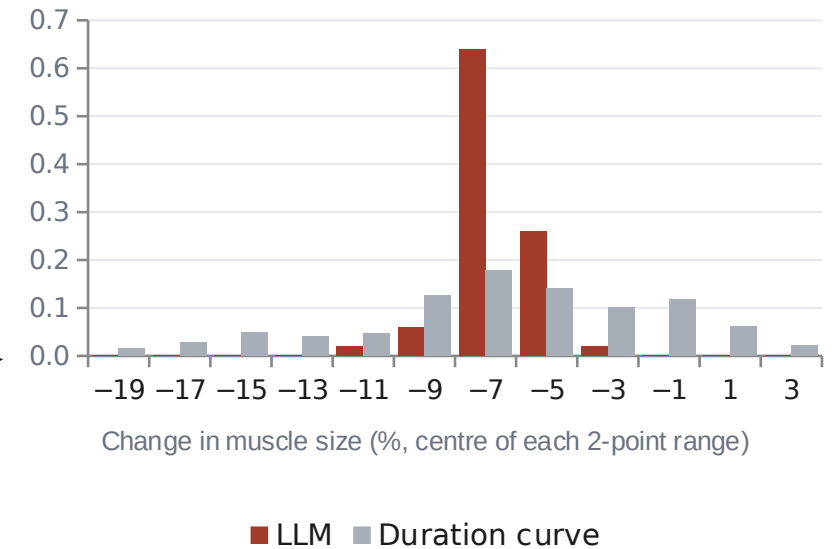
question

Which range will the change fall in? 19 options: below -30%, 2-point ranges up to +4%, +4% and above

Field names as sent; "campaign" means study.



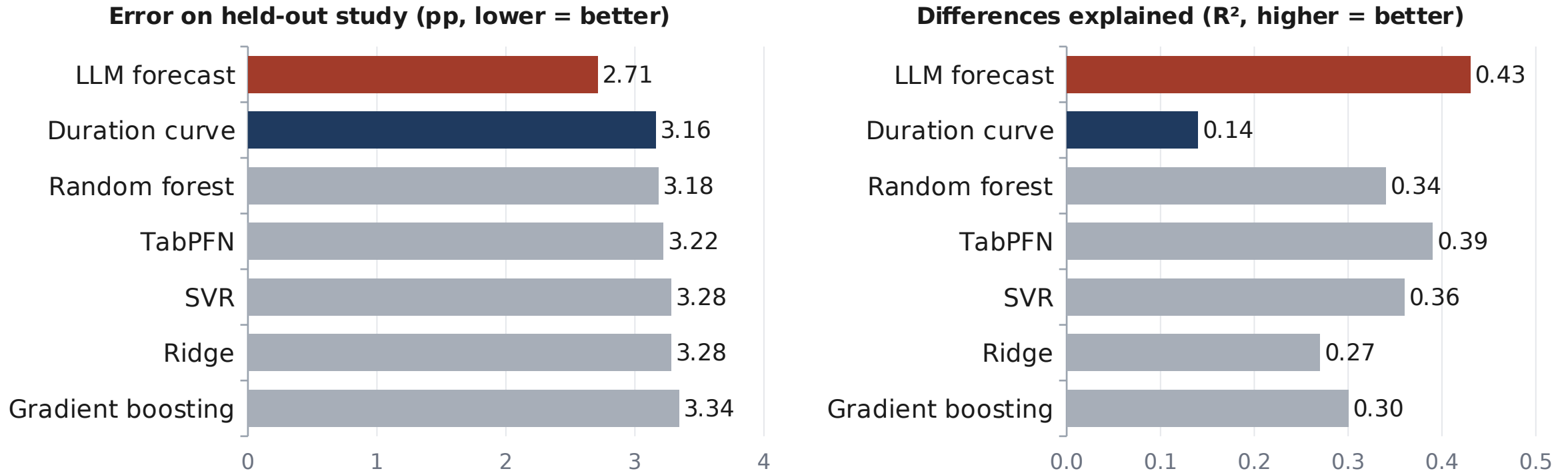
Answer: probability per range



- Our code turns the probabilities into one forecast, their weighted average: -6.6%
- Measured: -9.4%, an error of 2.8 pp. This is a typical row: half of the LLM's errors are smaller.

Never sent (checked on every request): the held-out study's measurements, and any paper, author or study name. All 430 answers are saved, so every result can be rebuilt without calling the model again.

The LLM forecast has the lowest error of all models



- Same measurements, scored the same way: error 2.71 vs 3.13 pp, so 0.42 pp smaller. Closer than the curve in 20 of 32 studies.
- 95% confidence interval of that difference: 0.13 to 0.73 pp. The whole range is above zero, so it is unlikely to be luck. No other model achieves this.

The curve shows 3.13 pp here, not 3.16, because the comparison scores it exactly as it scores the LLM.

Validity checks on the LLM forecast

Gain = how much smaller the LLM's error is than the curve's, in pp (full result: 0.42). Each check and its pass rule were fixed before running it.

Does it recognise published studies?

- Everything identifying a study removed: gain still 0.30 pp, interval above zero
- Asked to choose the study's name from a list of options: right 19% of the time, where guessing gives 8%

Does it use the other studies' data?

- Their values shuffled: worse than the curve (gain -1.25 pp)
- None of their data given: error 5.85 vs 3.13 pp

Is it sensitive to presentation?

- Other studies' rows in a different order: error changes 3%
- Answer ranges moved by 1 point: error changes 1%

Is it reproducible?

- 20 identical requests resent: forecasts differ by 0.17 pp on average, well below the 0.42 pp gain

Limitations of this result

- Added after the ML result was known, so it was not part of the original plan
- Overconfident: when it is 80% sure the value is in a range, it is there only 62% of the time. We therefore report only its single-number forecast.
- When also given a study's own earlier scans, it does not make use of them

Conclusions, limitations and further work

Conclusions

- With 32 independent studies, no ML model forecasts better than a curve based on days alone (best 3.18 vs 3.16 pp error).
- An LLM that reads each study's description and the other studies' data cuts the error to 2.71 pp (curve 3.13 pp), and this holds under every check.

Limitations

- Only 32 independent studies
- No data beyond day 119
- Mostly young men (347 of 425 measurements)
- Group averages, not individual people
- Bed rest imitates spaceflight but is not the same

Further work

- Fix the LLM's overconfident ranges
- Check results without the largest study, and with MRI scans only
- Extend the forecast to a 180-day mission, with an uncertainty range



Backup: is the dataset good enough?

Did you miss studies?

- Searched PubMed, Scopus, Web of Science and NASA reports from 2013 on (5,731 records), plus 9 older studies
- Not covered: Embase (no access) and 1,023 records not screened in time

Can MRI, CT, DXA and ultrasound be mixed?

- Each method gets its own term: DXA and ultrasound thickness show 5 to 6 pp less loss than CT; MRI volume is within 0.5 pp
- The MRI-only re-run is still to do

One study dominates the data

- MEDES 90-day: 40% of all rows, a quarter of the modelling data
- Scoring gives each study one vote; the re-run without it is still to do

How reliable is the extraction?

- Every value is traced to its paper and page, table or figure
- 674 of 742 rows high confidence; 19 read off figures; no independent second extraction

Backup: is the LLM gain real, and does it matter?

Is 0.42 pp worth anything?

- It is 13% lower error than the curve (3.13 to 2.71 pp)
- Small: we report it as a finding, not the headline

20 of 32 studies is close to a coin flip

- Counting wins only: $p = 0.22$ (sign test)*
- The size of the wins matters: mean gain 0.42 pp, 95% CI 0.13 to 0.73

Does it rest on a few studies?

- 3 long MRI studies (NASA SPRINT, Berlin BBR2-2, MEDES LTBR) give 56% of the gain; without them 0.20 pp, CI -0.01 to 0.43^*
- Only SPRINT is recognised, and with study details removed the LLM still wins there (3.79 vs 5.92 pp)

Did it read these papers in training? Was it added after the fact?

- Training exposure cannot be ruled out by any test; hiding study details and the naming test are the strongest checks available
- Added after the ML result, but every check and target was committed before the first answer

** Our calculation from results/forecast_predictions.csv; not in the report.*

Backup: why did ML fail, and was the comparison fair?

Did the ML models get a fair chance?

- Tuned inside each training fold, grouped by study; TabPFN needs no tuning
- All models saw the same inputs, folds and scoring

Why not add age, sex or countermeasure type?

- 32 studies leave room for about 3 study-level inputs (about 10 studies per input)
- Age and sex are fixed within a study (347 of 425 rows men-only), so they act as a study label; 126 countermeasure rows spread over 9 types

Is the duration curve a straw man?

- Ridge regression uses days, muscle and every other input: 3.28 vs 3.16 pp
- Knowing the muscle raises R^2 but not the error on a new study

Would a nearest-neighbour method match the LLM?

- Not tested. If the gain comes from picking similar rows, this is the obvious next control

Backup: can this be used to plan a Mars mission?

A Mars transit is about 180 days; the data stop at 119

- Only one study reaches 119 days
- A 180-day estimate is planned from the duration curve only, with an uncertainty range, labelled as an assumption

Which countermeasure works? What about individuals?

- Countermeasure groups lose 3.7 pp less (95% CI 1.5 to 5.9), all types pooled
- Every row is a group average; nothing about individual risk

Bed rest is not spaceflight

- Only 14 spaceflight rows, 8 of them one back muscle: too few to test transfer

Can anyone reproduce an LLM result?

- Fixed model version; all 430 answers saved, so every number rebuilds without the model
- Identical requests differ by 0.17 pp on average